

officials have found that key suspects took anaesthetic drugs (like Catamine, Propofol, Pentothal and Fortwin) to cheat narco tests, in case they were arrested. Regular use of the drugs made them immune to narco tests. This effectively puts a question mark on all 'revelations' made during narco tests so far.

Polygraph test

Polygraph examination is based on the assumption that there is an interaction between the mind and body. It is conducted by various components or the sensors of a polygraph machine, which are attached to the body of the person who is interrogated by the expert.

Polygraph machines are essentially biofeedback devices. The machine records the blood pressure, pulse rate, respiration, muscle movements and electrical resistance of the skin. Polygraph test is conducted in three phases: a pretest interview, chart recording and diagnosis.

Even with a trained and skillful administrator, polygraph technology is not altogether perfect at detecting lies. There is a lot of variability in how different people react when lying first of all. The act of being measured tends to produce anxiety, which creates the possibility of false positives. The worst part is that there are people out there who are so good at lying that they believe their own lies (or at least are in large part non-reactive) and thus don't perturb the polygraph.

Brain mapping

In the brain mapping method (P300), also called 'brain-wave fingerprinting,' the accused is first interviewed and interrogated to find out whether he is concealing any information. Then sensors are attached to his head and he is made to sit before a computer monitor. The accused is then shown certain images or made to hear certain sounds. The sensors monitor electrical activity in his brain and register P300 waves, which are generated only if the subject has connection with the stimulus, i.e.,

picture or sound. The subject is not asked any question here.

Functional magnetic resonance imaging

In functional magnetic resonance imaging (fMRI) experiments, the subject usually alternates between performing a mental task and resting (or undertaking an alternative task) while repeated images of his brain are rapidly captured. Areas of the brain in which there are strong correlations between the performance of the task and the MRI signal time course are then identified as having been involved in that task.

When the subject starts to perform the task, neuronal activity increases in the required parts of the brain. These areas require additional energy, which is provided by an increase in the regional blood supply. This increase in blood flow leads to an increase in the blood oxygenation level, which, in turn, changes the magnetic properties of the blood and hence the MRI signal.

Unfortunately, fMRI, as it is used today, has a major drawback: It measures blood flow, or haemodynamics, which is an indirect measure of neural cell activity. It turns out that haemodynamics basically introduces a delay of five seconds. So fMRI is not able to detect fast variation.

Since neurons typically fire at an interval of the order of milliseconds, current fMRI techniques provide only a rough estimate of what the brain is doing at any given moment. fMRI scans also have a relatively low spatial resolution, measuring activity in areas of 100 microns—a volume that typically contains 10,000 neurons, each with varying activation patterns. Efforts to fine-tune fMRI have focused on developing stronger magnets.

Calcium tracking

Researchers have found that tracking calcium—a key messenger in the brain—may be a more precise way of measuring neural activity than current

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